



Faculty of Engineering

School of Electrical Engineering and Telecommunications

ELEC 1111

Review

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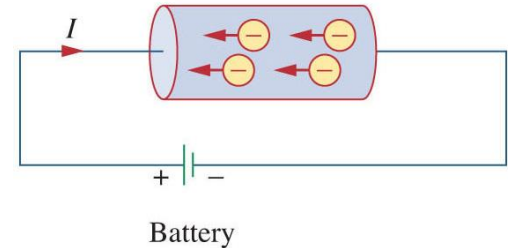
Final Exam Study Topics

- Topic 01: Introduction, Circuit Basic, Ohm's law, Sources and Diodes
- Topic 02: Kirchhoff's laws, Series & Parallel, Nodal and Mesh Analysis
- Topic 03: Circuit Theorems (Superposition, Thevenin, Norton, Source Transformation)
- Topic 04: Capacitors and Resistor-Capacitor (RC) Circuits
- Topic 05: Inductors and Resistor-Inductor (RL) Circuits
- Topic 06: Operational Amplifiers (Op Amps)
- Topic 07: AC Analysis I - Phasor and Impedance
- Topic 08: AC Analysis II - Circuit Theorems and AC Op Amps
- Topic 09: AC Power

Topic1 – Current, voltage and power

- Current is the rate of flow of electrons in a conductor.

$$i = \frac{dq}{dt}$$
$$q = \int_{t_0}^{t_1} i dt + q(t_0), \quad t_0 \leq t \leq t_1$$

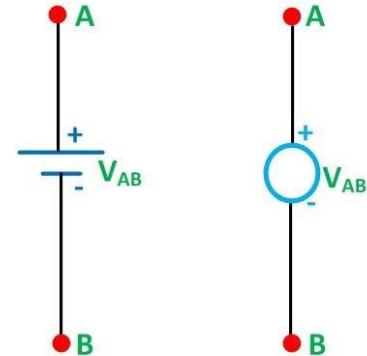


- Voltage is the potential difference between two points of the circuit (difference in charge between two locations):

$$v_{ab} = v_a - v_b = \frac{dw}{dq}$$

- Power and energy

$$p = \frac{dw}{dt} = vi$$
$$w = \int_{t_0}^t p dt = \int_{t_0}^t vi dt, \quad t_0 \leq t \leq t_1$$



Voltage and current are described by their values and direction/polarity.

Topic1 – Current, voltage and power

- **Passive sign convention**

- Power is **positive** if the current enters the **positive terminal**, $p = +vi$.
Power is **negative** if the current enters the **negative terminal**, $p = -vi$.
- Positive power is **absorbed/dissipated** by an element.
Negative power is **supplied/generated** by an element.



Note: Always use the sign to indicate whether power is generated or dissipated (even if you also clarify it with words).

- **Conservation of energy**

$$\sum p = 0$$

- **Sources**

- Voltage sources **generate or dissipate** power at a specified voltage with whatever current is required.
- Current sources **generate or dissipate** power at a specified current with whatever voltage is required.
- Sources can be ideal/real, dependent/independent.

Topic 1 - Resistance

- Resistance R is a physical property of materials that impedes the flow of charge.
- Ohm's Law establishes the relationship between the voltage across a resistor and the current through it:

$$v = Ri$$

- Short circuit: $R = 0 \Omega \rightarrow v = Ri = 0$.

Open circuit: $R = \infty \Omega \rightarrow i = \frac{v}{R} = 0$.



- Power dissipation on resistor:

$$p = vi = Ri^2 = \frac{v^2}{R}$$

- Resistors in series: $R_{eq} = R_1 + R_2 + \dots + R_N$

- Resistors in parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}$

Topic 2 – Kirchhoff's laws, voltage/current division

- Kirchhoff's Current Law (KCL):

- The sum of all currents entering and leaving a node is zero:

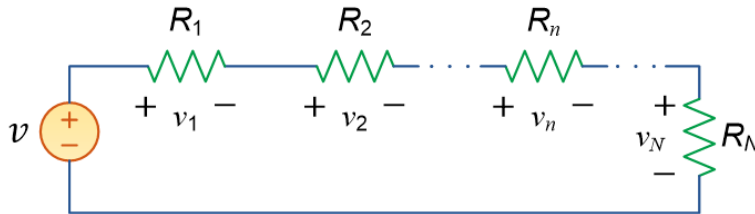
$$\sum_{n=1}^N i_n = 0$$

- Kirchhoff's Voltage Law (KVL):

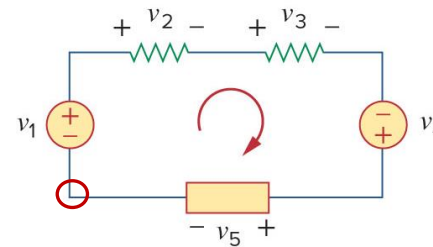
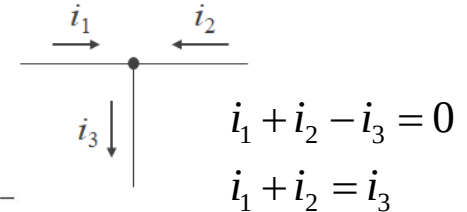
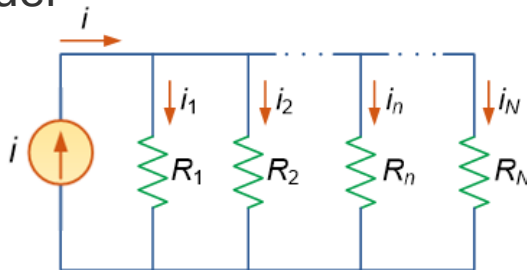
- The sum of all voltages in a loop is zero:

$$\sum_{m=1}^M v_m = 0$$

- Voltage divider



- Current divider



$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

$$v_n = v \frac{R_n}{R_1 + R_2 + \dots + R_N} = v \frac{R_n}{R_{eq}}$$

$$i_n = i \frac{R_1 \parallel R_2 \parallel \dots \parallel R_N}{R_n} = i \frac{R_{eq}}{R_n}$$

Topic 2 - Nodal Analysis

Nodal analysis



Before starting working:

- Simplify and label your circuit.
- Check the number of simultaneous equations (i.e. think about the most appropriate method).

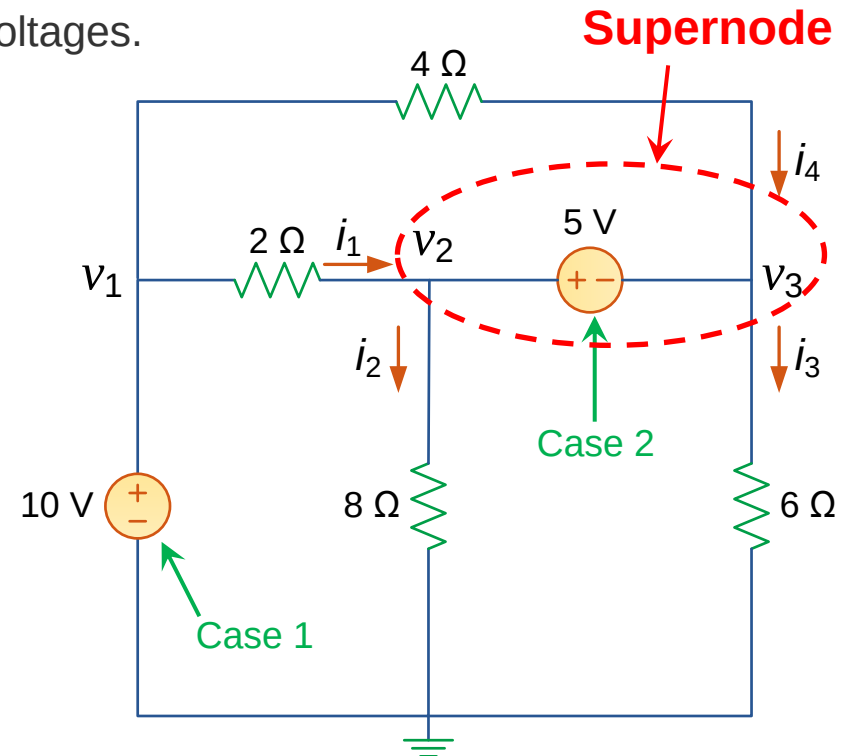
- Three steps:
 - Set the reference node and node voltages.
 - Apply KCL and use Ohm's law to write currents in terms of node voltages.
 - Solve the set of linear equations for node voltages.

- Nodal analysis with voltage source:
 - Case 1: Voltage source is connected to the reference node.

$$v_k = v_{\text{source}}$$

- Case 2: Voltage source is between two non-reference nodes.
 - Form a supernode by enclosing voltage source and any parallel element with it.
 - Write the constraint equation relating the node voltages inside the supernode.

$$v_a - v_b = v_{\text{source}}$$



Topic 2 - Mesh analysis

Mesh analysis

- Three steps:
 - Set the mesh currents with arbitrary direction (generally clockwise).
 - Apply KVL and use Ohm's law to write voltages in terms of mesh currents.
 - Solve set of linear equations for mesh currents.

- Mesh analysis with current source:

- Case 1: Current source is only in one mesh.

$$i_k = i_{\text{source}}$$

- Case 2: Current source is shared between two meshes.

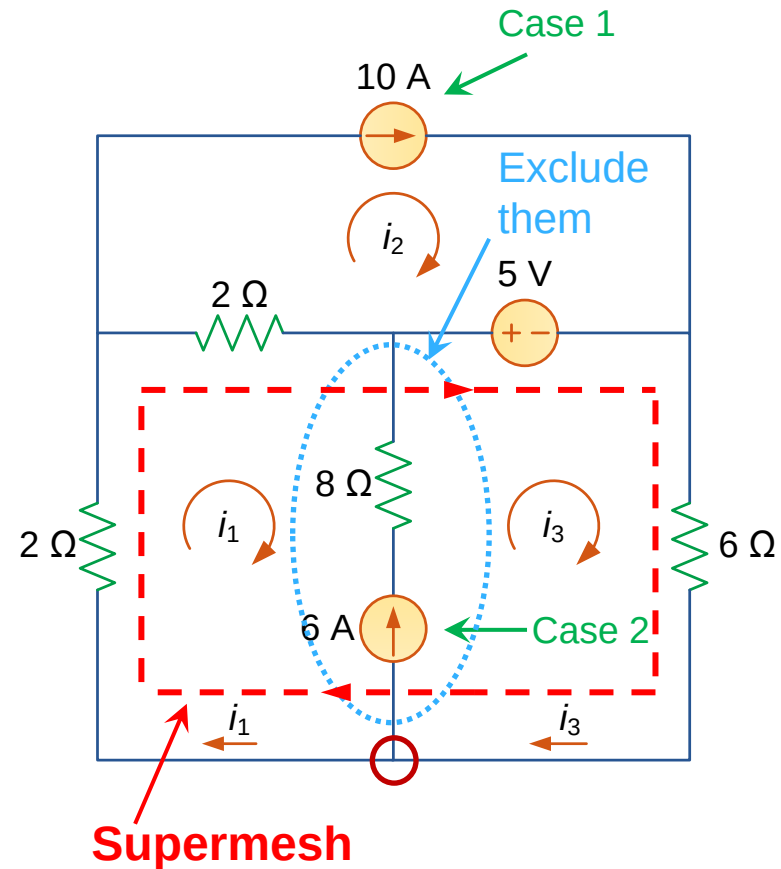
- Form a supermesh by excluding the current source and any series element with it.
- Write the constraint equation relating the mesh currents inside the supermesh.

$$i_a - i_b = i_{\text{source}}$$



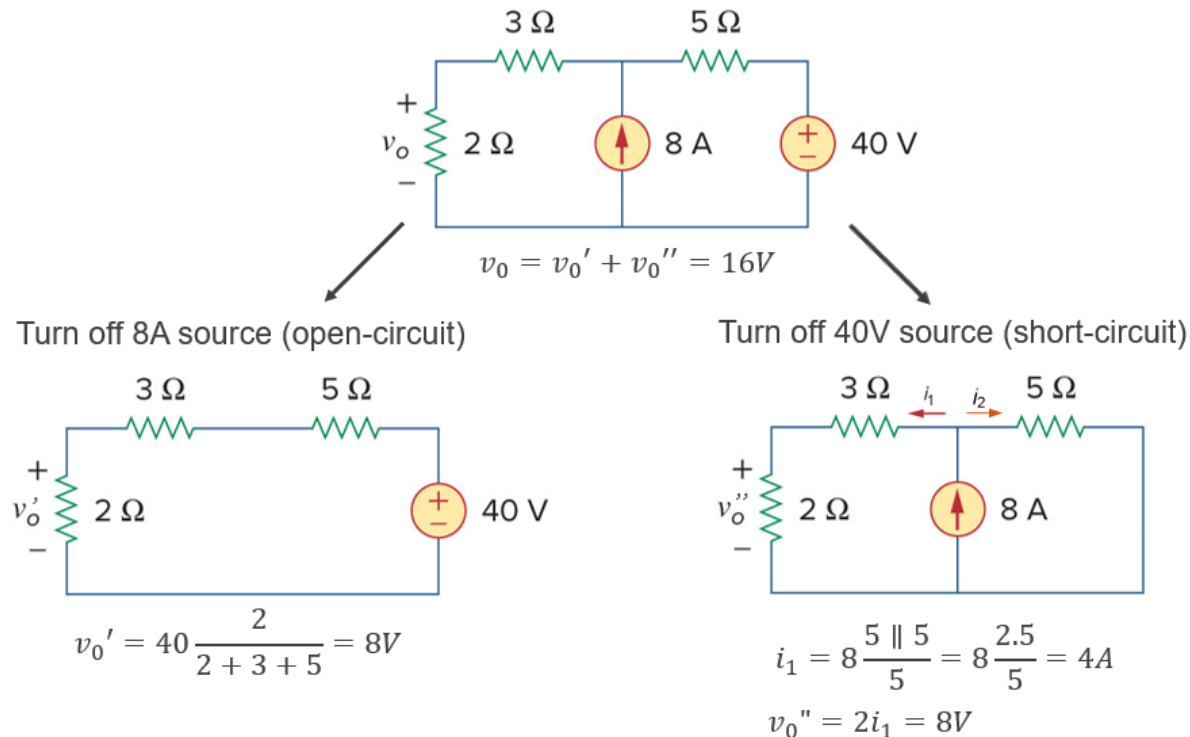
Before starting working:

- Simplify and label your circuit.
- Check the number of simultaneous equations (i.e. think about the most appropriate method).



Topic 3 - Superposition

- Superposition principle in linear circuits (circuit analysis).
 - The response of a linear circuit (a voltage or current of an element) with multiple sources is the **algebraic sum** of individual responses **due to each independent source** when the rest are **turned off**.
 - **Dependent sources** are **left intact** during all calculations.
- Example with independent sources:

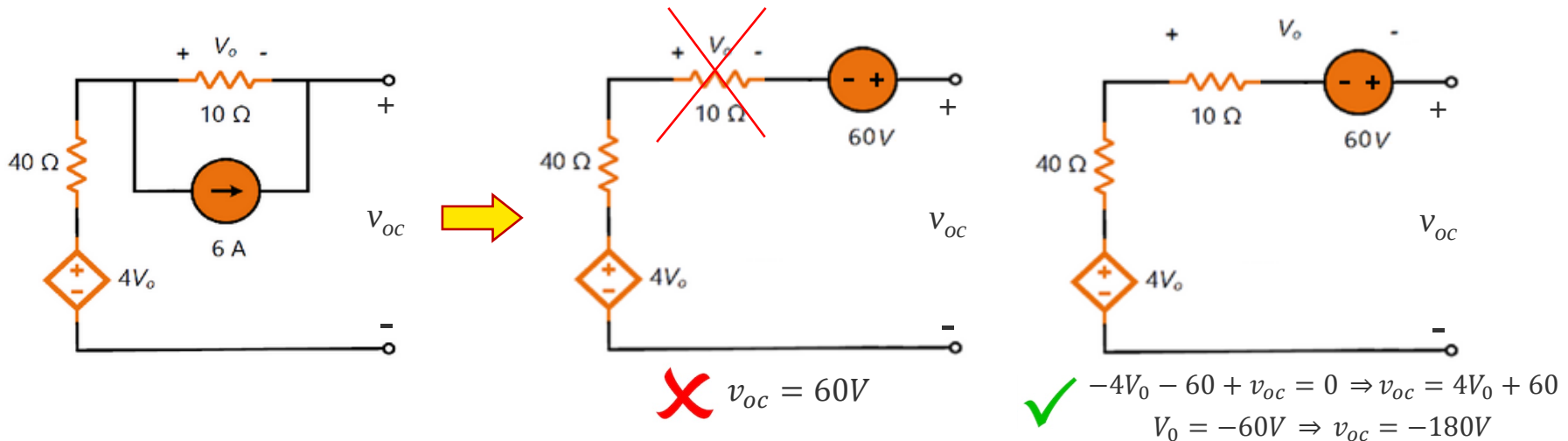


Topic 3 – Source transformation

- Source transformation (circuit analysis/simplification).
 - Consists in replacing a **voltage source** v_s (independent or dependent) in **series** with a **resistor** with a **current source** i_s (independent or dependent, respectively) in **parallel** with another resistor at the **same terminal** or vice versa.

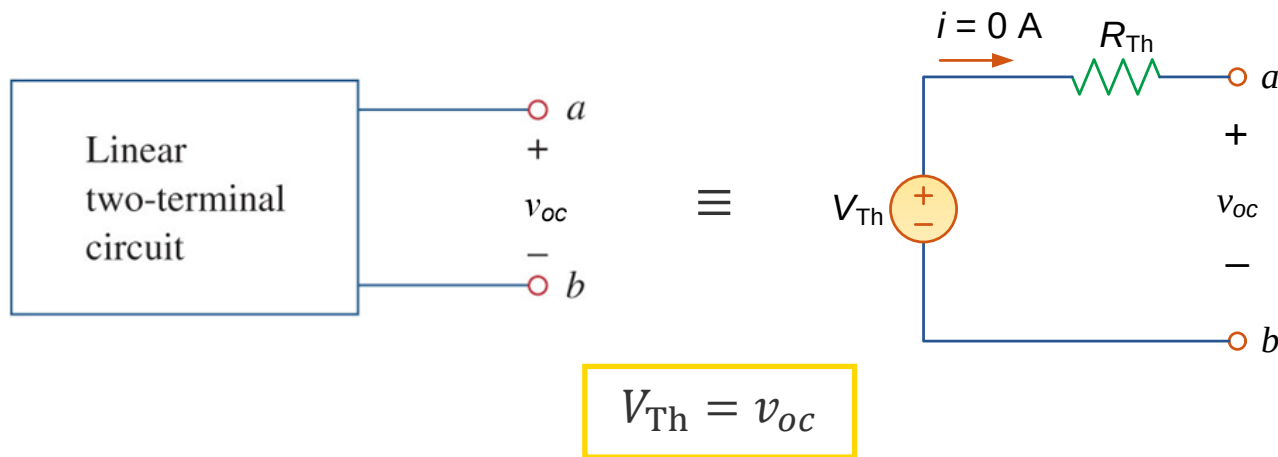
- $i_s = \frac{v_s}{R}$ or $v_s = Ri_s$.
- The transformed resistors are the **same**.

- Source transformation can also be applied to dependent sources, provided that the dependent variable is handled carefully.
 - Example of source transformation with dependent source (calculate v_{oc}):



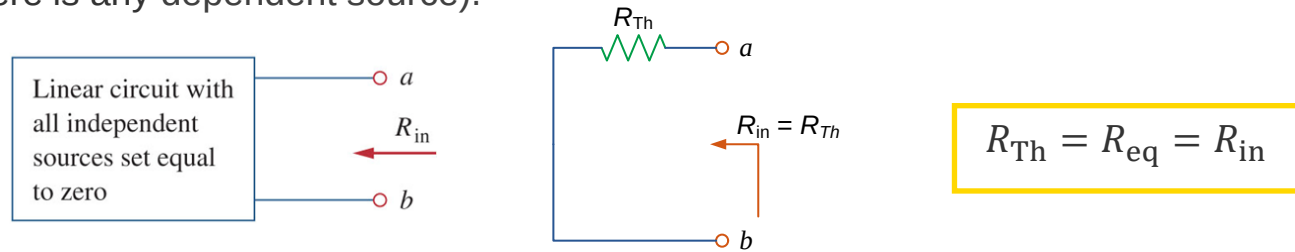
Topic 3 – Thevenin's theorem

- Thevenin's theorem (circuit analysis/simplification).
 - A linear circuit can be **replaced** (modelled) with a **voltage source** V_{Th} in **series** with a **resistor** R_{Th} from a given terminal.
 - V_{Th} is equal to the **open-circuit voltage** across the terminals ($V_{Th} = v_{oc}$).

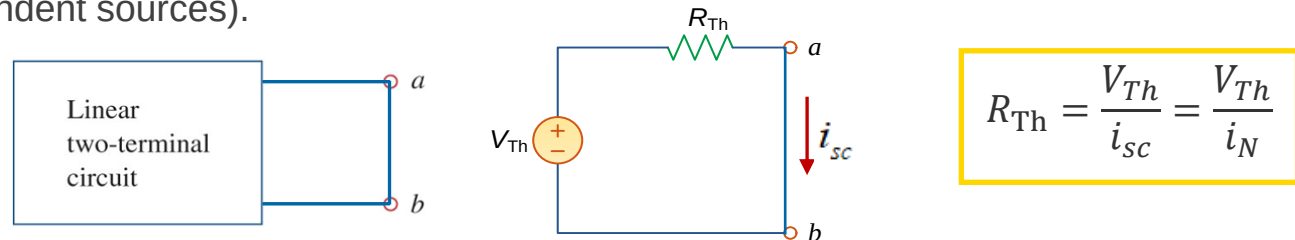


Topic 3 – Thevenin's theorem

- R_{Th} can be obtained using different methods:
 1. Input **resistance** measured at the terminal pair when **all independent sources are turned off** (not valid if there is any dependent source).

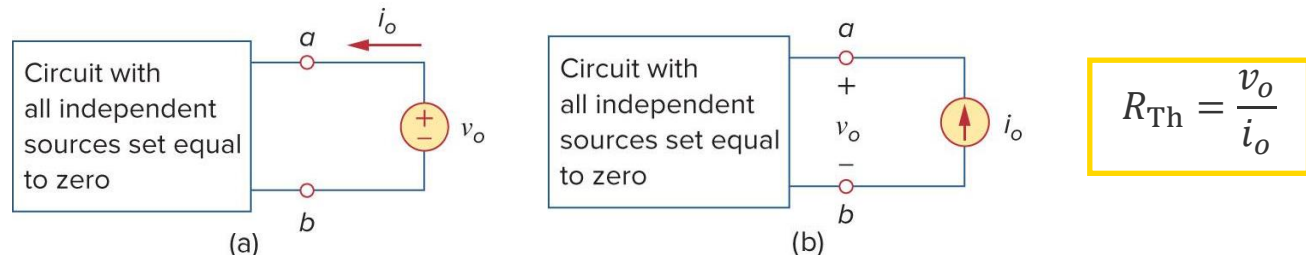


2. Ratio of the open-circuit voltage to the short-circuit current at the terminal pair (not valid if there are only dependent sources).



3. Turn off independent sources and (a) attach a voltage source v_o to the terminals a - b and find the resulting current i_o , or (b) attach a current source i_o and find the resulting voltage v_o . For simplicity: $v_o = 1$ V or $i_o = 1$ A (valid always).

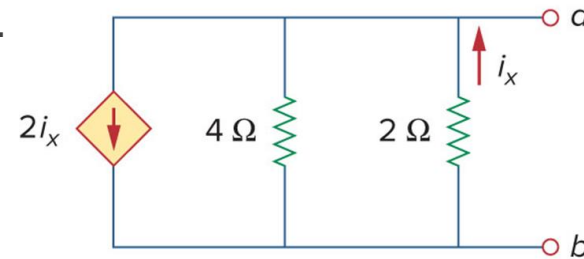
⚠ Note: If there are only dependent sources in the circuit (dead network), it is possible for R_{Th} to be negative ($R_{Th} < 0$). This implies that the circuit is supplying power.



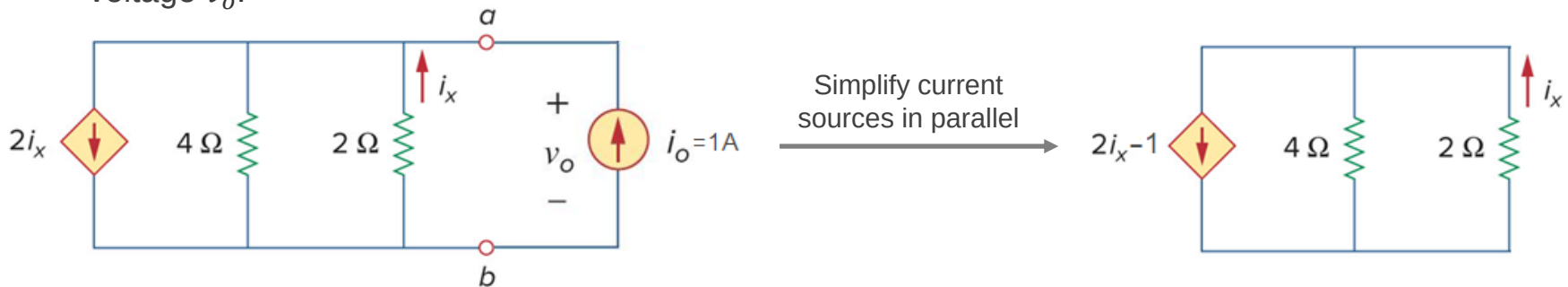
Topic 3 – Thevenin's theorem

- Example of circuit with dependent source only.

Find the Thevenin equivalent at terminals a - b .



- $V_{Th} = v_{oc} = 0$ (since there are no independent sources in the circuit).
- To calculate R_{Th} we attach a current source i_o ($i_o = 1$ A) to the terminals and find the resulting voltage v_o .



Use current division to find i_x :
$$i_x = (2i_x - 1) \frac{4 \parallel 2}{2} = (2i_x - 1) \frac{\frac{1}{\frac{1}{4} + \frac{1}{2}}}{2} = (2i_x - 1) \frac{4}{4+2} = \frac{2}{3}(2i_x - 1)$$

$$i_x - \frac{4}{3}i_x + \frac{2}{3} = 0; \quad 3i_x - 4i_x + 2 = 0 \rightarrow i_x = 2A$$

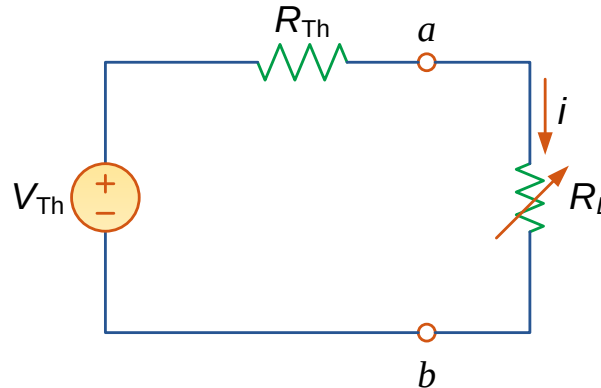
$$v_o = -2i_x = -4V \rightarrow R_{Th} = \frac{v_o}{i_o} = \frac{-4}{1} = -4\Omega$$

Topic 3 – Thevenin's and Norton's theorems

- Norton's theorem.
 - A linear circuit can be **replaced** (modelled) with a **current source** I_N in **parallel** with a **resistor** R_N from a given terminal.
 - I_N is equal to the **short-circuit current** at the terminals ($I_N = i_{sc}$).
 - R_N is the **same** as R_{Th} .
- Thevenin-Norton transformation is exactly the **same** as **source transformation**.
 - $V_{Th} = R_N I_N$ or $I_N = \frac{V_{Th}}{R_{Th}}$
 - $R_{Th} = R_N = \frac{v_{oc}}{i_{sc}}$

Topic 3 – Maximum power transfer

- In many practical applications, a circuit is designed to provide maximum power to a load.
- Using the Thevenin equivalent circuit we can find the maximum power that can be transferred to a load based on fixed V_{Th} and R_{Th} and variable R_L .



- **Maximum power** is transferred to the load **when the load resistance R_L is equal to the Thevenin resistance R_{Th}** as seen from the load terminals.

$$\boxed{R_L = R_{Th}} \Rightarrow \begin{cases} \boxed{p_{\max} = \frac{V_{Th}^2}{4R_{Th}}} \\ \boxed{v_L = \frac{V_{Th}}{2}} \end{cases}$$

Topic 4 - Capacitors

- A capacitor is a circuit element that stores energy in its electric field.
- The ratio of voltage to charge across a capacitor is its *capacitance* (F).

$$q = Cv$$

- Current in a capacitor is proportional to the time rate of change of its voltage.

$$i = C \frac{dv}{dt}$$

$$v(t) = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$$

- Energy stored in a capacitor is proportional to the square of its voltage.

$$w_c = \frac{1}{2} C v^2$$

- **Capacitor acts as an open circuit to DC voltage.**
- Parallel combination of capacitors is similar to series resistors.

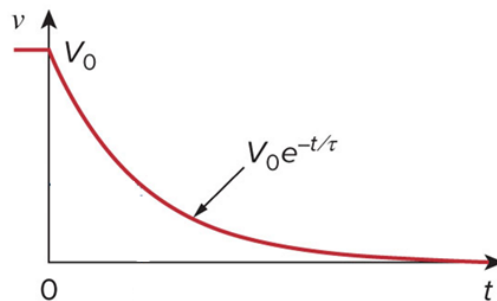
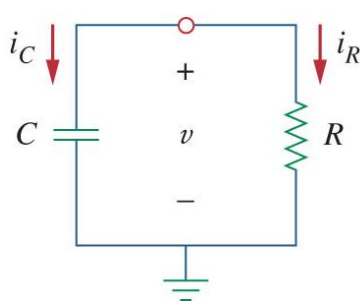
$$C_{eq} = C_1 + C_2 + \dots + C_N$$

- Series combination of capacitors is similar to parallel resistors.

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \dots + \frac{1}{C_N}$$

Topic 4 – Natural response of RC circuits

- Natural response of RC circuits
 - Behaviour of the circuit (in terms of voltage or current) due to initial energy stored.
 - If the capacitor has a initial voltage $v(0) = V_0$, the natural response of the RC circuit is:



$$v(t) = V_0 e^{-\frac{t}{RC}} = V_0 e^{-\frac{t}{\tau}}$$

- The speed at which the voltage decays is given by the *time constant*, τ :
 - Time constant is the time required for the response to decay to a factor of $1/e$ or **36.8%** of its **initial value** or to reach **63.2%** of its **final value**.
 - For an RC circuit $\tau = RC$.
 - After **5 time constant**, 5τ , the capacitor voltage is considered to reach its final value.
 - The higher R and C, the longer it would take for the capacitor to charge or discharge.
 - The resistance for the time constant is the **Thevenin equivalent resistance** as seen from the capacitor terminals.

$$\tau = R_{Th}C$$

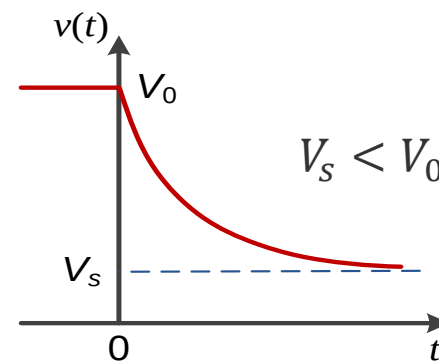
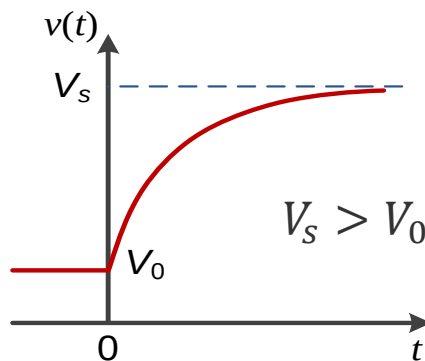
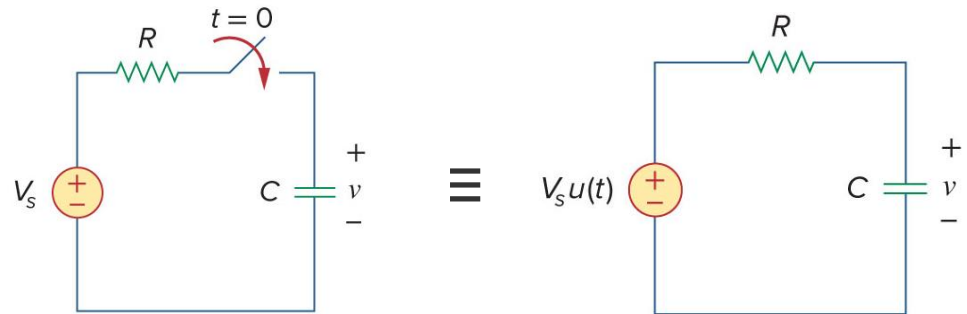
Topic 4 - Step response of RC circuits

- Step response of RC circuits
 - The unit step function can be used in electric circuits to model switching.

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

- The step response is the response to a sudden change in the input sources.

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-\frac{t}{RC}}, & t > 0 \end{cases}$$



Topic 4 – Step response of RC circuits

- Step response of RC circuits
 - The capacitor voltage over time is obtained as an exponential function:

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-\frac{t}{\tau}}, & t > 0 \end{cases}$$

- If the initial voltage $v(0) = 0$ V , the response is known as **forced response**.
- The step response with non-zero initial condition is known as **complete response**.
 - It can be described as the sum of **transient** and **steady state responses**.
- The solution to the step response of RC circuits can be given as follows:

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\frac{t}{\tau}}, \quad t > 0$$

- $v(0)$: **Initial voltage** at $t = 0$.
- $v(\infty)$: **Final or steady-state value** at $t \rightarrow \infty$.
- $\tau = R_{Th_\infty}C$: Time constant at $t \rightarrow \infty$.



Note: To calculate the solution to the natural response of RC circuits, you can substitute $v(\infty) = 0$ in the previous formula (note that for natural response, the capacitor is completely discharged after 5τ ; hence, $v(\infty) = 0$).

Topic 4 - Time shift in step response of RC circuits



Note that if one switch changes position at time $t = t_0$ instead of $t = 0$, there is a time delay in the response which can be expressed as time shift in the equation.

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{\frac{-(t-t_0)}{\tau}}, \quad t > t_0$$

Topic 5 - Inductors

- An inductor is a circuit element that stores energy in its magnetic field.
- Inductance L is the property of inductors by which they oppose to changes in the current flowing through them. It is measured in henry (H).
- Voltage in an inductor is proportional to the time rate of change of its current.

$$v = L \frac{di}{dt}$$

$$i(t) = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$$

- Energy stored in an inductor is proportional to the square of its current.

$$w_L = \frac{1}{2} L i^2$$

- **Inductor acts as a short circuit to DC voltage.**
- Series combination of inductors is similar to series resistors.

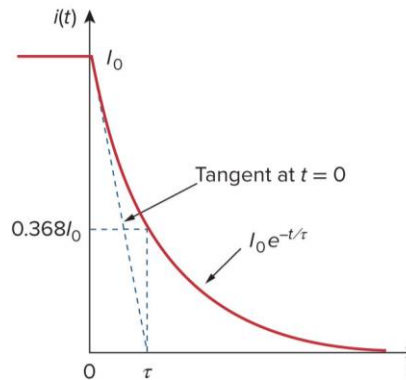
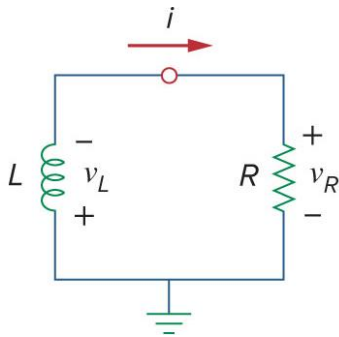
$$L_{eq} = L_1 + L_2 + \cdots + L_N$$

- Parallel combination of inductors is similar to parallel resistors.

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} \cdots + \frac{1}{L_N}$$

Topic 5 – Natural response of RL circuits

- Natural response of RL circuits
 - Behaviour of the circuit (in terms of voltage or current) due to initial energy stored.
 - If the inductor has an initial current $i(0) = I_0$, the natural response of the RL circuit is:



$$i(t) = I_0 e^{-\frac{R}{L}t} = I_0 e^{-\frac{t}{\tau}}$$

- The speed at which the current decays is given by the *time constant*, τ :
 - Time constant is the time required for the response to decay to a factor of $1/e$ or **36.8%** of its **initial value** or to reach **63.2%** of its **final value**.
 - After **5 time constant**, 5τ , the inductor current is considered to reach its final value.
 - For an RL circuit $\tau = \frac{L}{R}$.
 - The resistance for the time constant is the **Thevenin equivalent resistance** as seen from the inductor terminals.

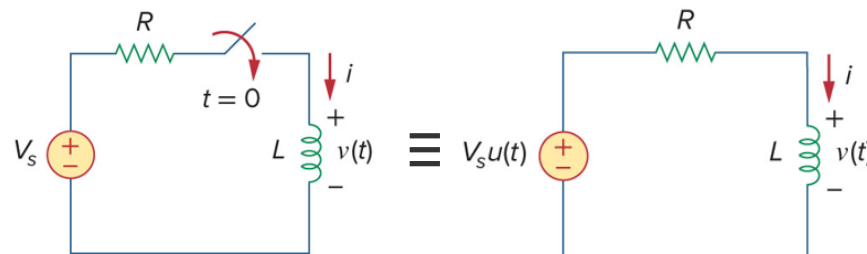
$$\tau = \frac{L}{R_{Th}}$$

Topic 5 – Step response of RL circuits

- Step response of RL circuits

- The unit step function can be used in electric circuits to model switching.

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

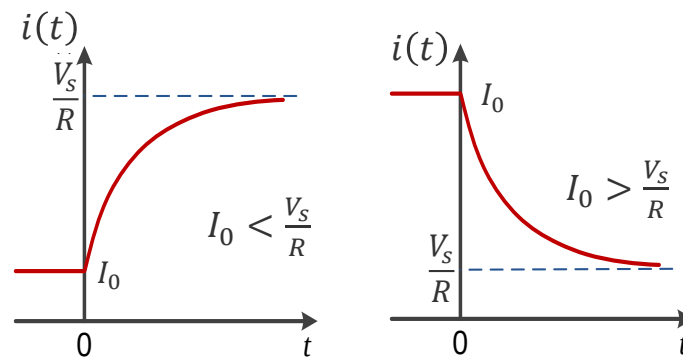


- The inductor current over time is obtained as: $i(t) = \begin{cases} I_0, & t < 0 \\ \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-\frac{t}{\tau}}, & t > 0 \end{cases}$
- If the initial current $i(0) = 0$ A, the response is known as **forced response**.
- The step response with non-zero initial condition is known as **complete response**.
 - It can be described as the sum of **transient** and **steady state responses**.
- The solution to the step response of RL circuits can be given as follows:

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}, \quad t > 0$$

- $i(0)$: **Initial current** at $t = 0$.
- $i(\infty)$: **Final or steady-state value** at $t \rightarrow \infty$.
- $\tau = \frac{L}{R_{Th_\infty}}$: Time constant at $t \rightarrow \infty$.

! **Note:** use $i(\infty) = 0$ for natural response.

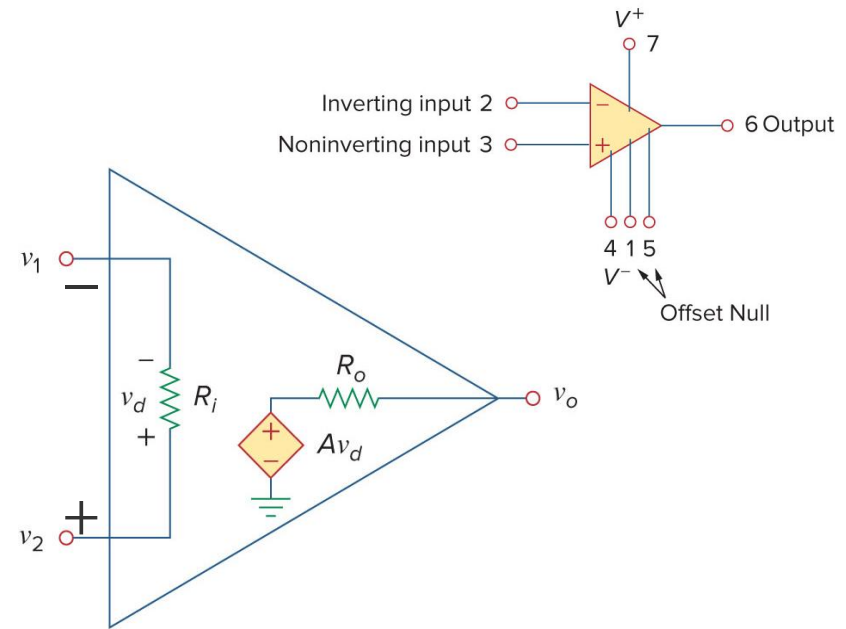


Topic 6 – Op Amp equivalent circuit model

- Operational Amplifiers (Op Amps) are **active** elements.
 - Op Amp circuits are designed to perform **mathematical operations** on input signals. They are manufactured in the form of Integrated circuits (ICs).
- An Op Amp is modelled from its input terminals by an input resistor R_i , and from its output terminal by a voltage-controlled voltage source in series with an output resistor R_o .
- The output voltage v_o is given by:

$$v_o = Av_d = A(v_2 - v_1)$$

! **Note:** This equation describes the **internal** behaviour of the Op Amp. When analyzing an Op Amp circuit, we are not interested in the internal behavior or the open-loop gain. We want to know what happens when we connect different circuit elements (resistors, inductors or capacitors) to the inputs and/or output, and we focus in the close-loop gain. Steps for analysis are explained in the next slide.



- A : Open-loop voltage gain
- v_d : Differential input voltage
- v_1 : Inverting terminal voltage to the ground
- v_2 : Non-inverting terminal voltage to the ground
- R_o : Thevenin equivalent resistance seen at output terminals
- R_i : Thevenin equivalent resistance seen at the input terminals

Topic 6 – Op Amp circuit analysis

- An **Ideal Op Amp** has the following properties:
 - Open-loop gain close to infinity, $A \simeq \infty$.
 - Input resistance close to, infinity $R_i \simeq \infty \Omega$ (open circuit).
 - Output resistance close to zero, $R_o \simeq 0 \Omega$ (short circuit).
 - The **currents** into both terminals are **zero**.

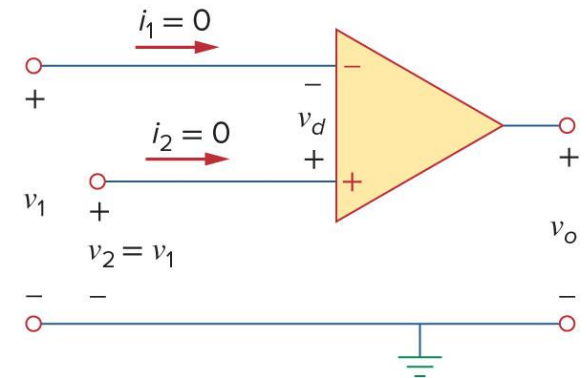
$$i_1 = i_2 = 0$$

- The **voltage** across the input terminals is **zero** (if there is negative feedback).

$$v_d = v_2 - v_1 = 0 \quad \text{or} \quad v_1 = v_2$$

- Use **Nodal analysis** to solve and analyse Op Amp circuits.
 - KCL at input nodes to calculate v_o .
 - Once output voltage v_o is found, use KCL at output node to find output current i_o (if needed).
 - Gain of the circuit = ratio of output to the input (v_o / v_i).
- Op Amp circuits should be operated in the linear region to avoid saturation of the output (output should not exceed the voltage of power supply - Positive power supply V^+ (pin 7) and negative power supply V^- (pin 4), more commonly known as $\pm V_{CC}$).

$$-V_{CC} \leq v_o \leq V_{CC}$$

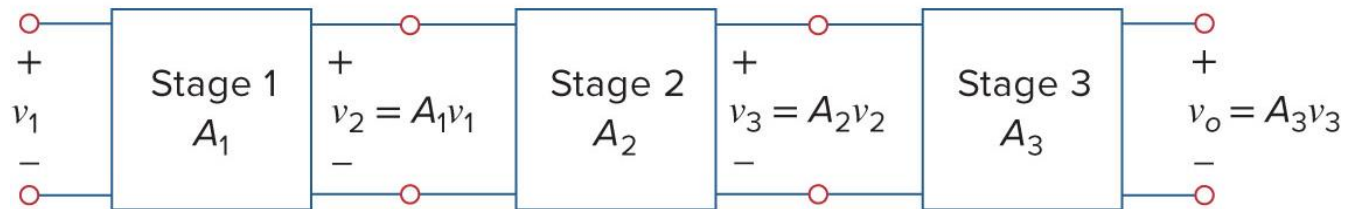


Topic 6. Cascade of Op Amp stages

- It is common to use multiple Op Amp circuits chained together to increase the overall gain of an amplifier.
 - Due to input and output impedances of the ideal Op Amps, stages can be connected together without affecting the performance of each other (**no loading effect**).

 **Note:** This idea of “no loading effect” also applies when the Op Amps are connected to other types of circuits.

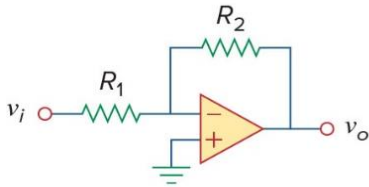
- This head to tail configuration is called “cascading”.
- Each amplifier is then called a “stage”.



- The gain of a series of amplifiers connected in cascade is the product of the individual gains:

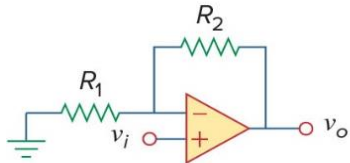
$$A = A_1 \times A_2 \times A_3$$

Topic 6 – Useful Op Amp circuits



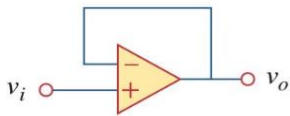
Inverting amplifier

$$v_o = -\frac{R_2}{R_1}v_i$$



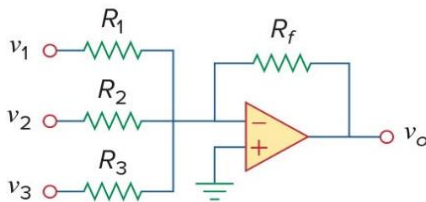
Noninverting amplifier

$$v_o = \left(1 + \frac{R_2}{R_1}\right)v_i$$



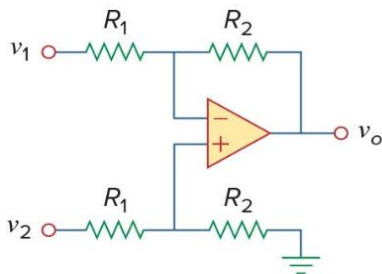
Voltage follower

$$v_o = v_i$$



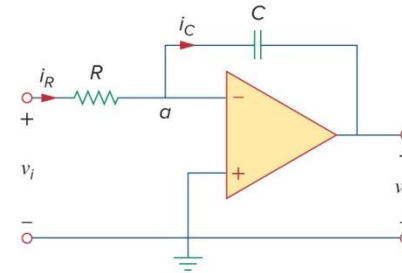
Summer

$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$$



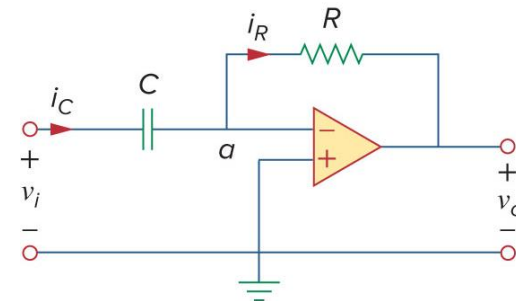
Difference amplifier

$$v_o = \frac{R_2}{R_1}(v_2 - v_1)$$



Integrator

$$v_o(t) = -\frac{1}{RC} \int_0^t v_i(\tau) d\tau$$



Differentiator

$$v_o(t) = -RC \frac{dv_i}{dt}$$

Topic 7 – AC circuits and sinusoids

- AC circuits are operated by sinusoidal sources.
- A **sinusoid** is a signal (voltage or current) in the form of sine or cosine.

$$v(t) = V_m \cos(\omega t + \phi)$$

V_m : Amplitude.

$\omega = 2\pi f$: Angular frequency (rad/s).

$(\omega t + \phi)$: Argument of the sinusoid.

ϕ : Phase (in degrees or radians).

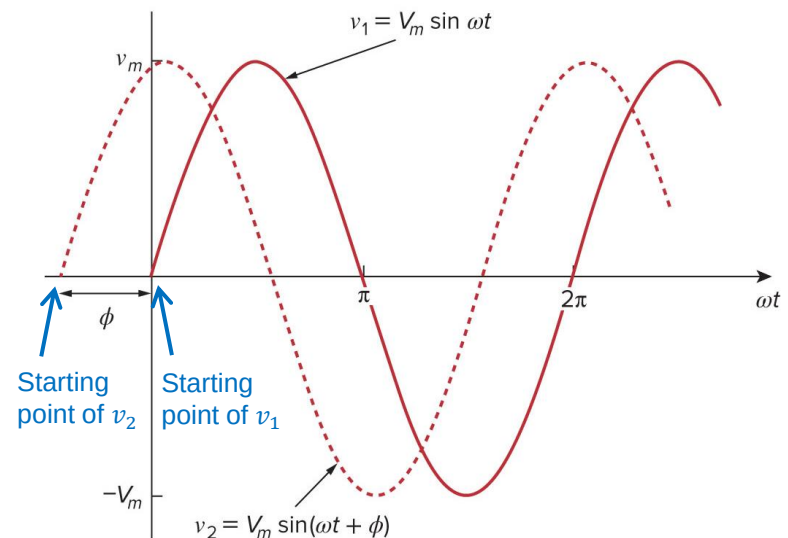
- Given 2 sinusoids with different phases:

$$v_1(t) = V_m \sin(\omega t)$$

$$v_2(t) = V_m \sin(\omega t + \phi)$$

- If $\phi \neq 0$, v_1 and v_2 are **out of phase**.
- If $\phi = 0$, v_1 and v_2 are **in phase** (they reach maxima and minima at the same time).
- It can be assumed that v_1 starts at a later time ($t = 0$ s) compared to v_2 which begins earlier at $t = -\frac{\phi}{\omega}$ s. Thus,

v_1 is said to **lag** v_2
 v_2 is said to **lead** v_1



Topic 7 - Phasors



$$\begin{aligned}\cos(\omega t - 180^\circ) &= -\cos \omega t \\ \cos(\omega t - 90^\circ) &= \sin(\omega t)\end{aligned}$$

- **Phasor** is a **complex number** that represents the **amplitude** and **phase** of a **sinusoid**.
 - For a given sinusoid in cosine form, $v(t) = V_m \cos(\omega t + \phi)$, the phasor is a complex number by suppressing the time factor $e^{j\omega t}$, and taking the amplitude V_m and the phase ϕ .
 - To obtain the time-domain representation of a given phasor V , use a cosine function with the same magnitude as the phasor and the argument ωt plus the phase of the phasor.

Time-domain representation

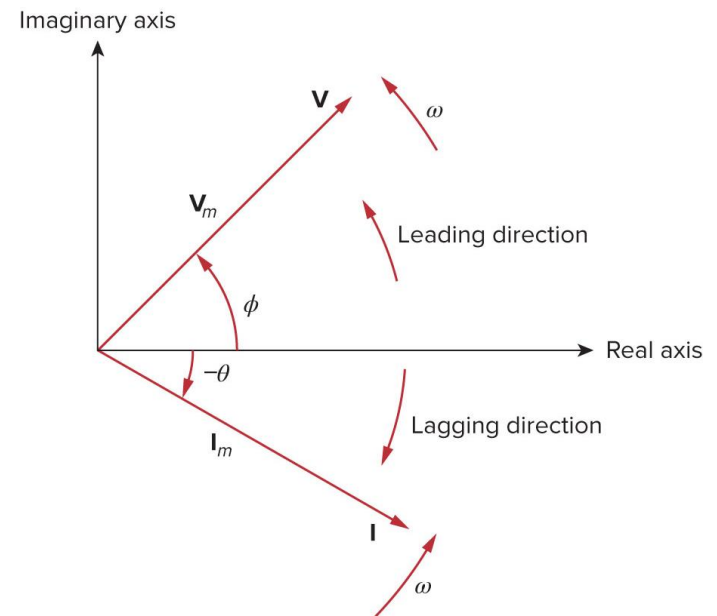
$$v(t) = V_m \cos(\omega t + \phi)$$



$$\mathbf{V} = V_m \angle \phi = V_m e^{j\phi}$$

Phasor-domain representation

- A phasor can be expressed as a **vector** with a **magnitude** and a **phase** (direction).
- **Sketching** the phasors in a Cartesian coordinates or complex plane is called **phasor diagram**.
 - The convention for measuring angle/phase is from positive real axis rotating counter clockwise.



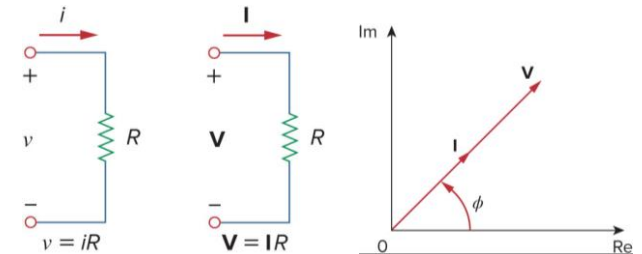
Phasor diagram showing
 $\mathbf{V} = V_m \angle \phi$ and $\mathbf{I} = I_m \angle -\theta$

Topic 7 – V-I relationship

- Circuit elements have a fixed relationship between voltage and current phasors.
- Given $v(t) = V_m \cos(\omega t + \phi_v) \Leftrightarrow \mathbf{V} = V_m \angle \phi_v$ as the voltage across an element and $i(t) = I_m \cos(\omega t + \phi_i) \Leftrightarrow \mathbf{I} = I_m \angle \phi_i$ as the current through the element:

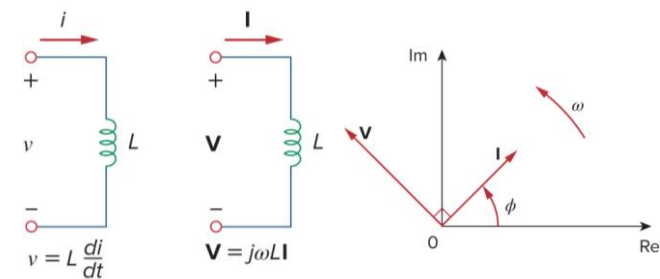
- For resistor R , voltage and current are **in phase**:

$$v = Ri \Rightarrow \mathbf{V} = R\mathbf{I} = RI_m \angle \phi_i = V_m \angle \phi_v \Rightarrow \phi_v = \phi_i$$



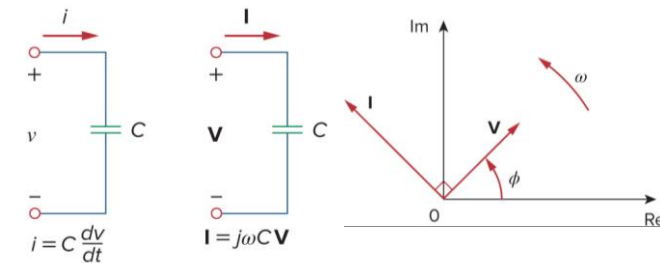
- For inductor L , current **lags** voltage by 90° :

$$v = L \frac{di}{dt} \Rightarrow \mathbf{V} = j\omega L \mathbf{I} = \omega L I_m \angle (\phi_i + 90^\circ) = V_m \angle \phi_v \Rightarrow \phi_v = \phi_i + 90^\circ$$



- For capacitor C , current **leads** voltage by 90° :

$$v = \frac{1}{C} \int i dt \Rightarrow \mathbf{V} = \frac{1}{j\omega C} \mathbf{I} = \frac{I_m}{\omega C} \angle (\phi_i - 90^\circ) = V_m \angle \phi_v \Rightarrow \phi_v = \phi_i - 90^\circ$$



Topic 7 – Impedance

- **Impedance Z** of a circuit is the ratio of the phasor voltage across it to the phasor current through it.

$$Z = \frac{V}{I} = R(\omega) + jX(\omega) \Omega, \quad R: \text{Resistance}, X: \text{Reactance}.$$

- Admittance Y is the reciprocal of impedance.

$$Y = \frac{1}{Z} = \frac{I}{V} = G(\omega) + jB(\omega) S, \quad G: \text{Conductance}, B: \text{Susceptance}.$$

- Impedances of circuit elements:

- For resistor R : $Z_R = R$
- For inductor L : $Z_L = j\omega L$
- For capacitor C : $Z_C = 1/j\omega C = -j/\omega C$

- Impedances are **combined** in series and parallel in the **same way** as **resistances** in series and parallel.

$$Z_{\text{series}} = Z_1 + Z_2$$

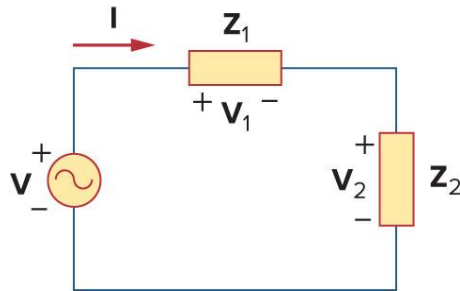
$$Z_{\text{parallel}} = \frac{1}{\frac{1}{Z_1} + \frac{1}{Z_2}} = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

Topic 7 – Ohm's & Kirchhoff's laws, voltage/current division

- Basic circuit laws (**Ohm's** and **Kirchhoff's**) apply to AC circuits in the same manner as DC circuits, as well as voltage and current **divisions**.

- $V = ZI$ (Ohm's law)
- $\sum I_k = 0$ (KCL) and $\sum V_k = 0$ (KVL)

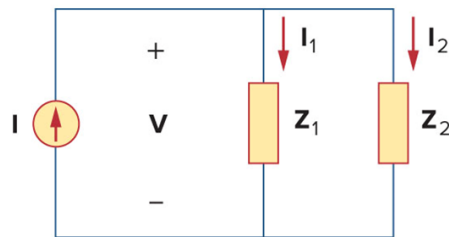
- Voltage division for impedances works the same way as voltage division for resistors.



$$V_1 = \frac{Z_1}{Z_1 + Z_2} V$$

$$V_2 = \frac{Z_2}{Z_1 + Z_2} V$$

- Current division for impedances works the same way as current division for resistors.



$$I_1 = \frac{Z_2}{Z_1 + Z_2} I$$

$$I_2 = \frac{Z_1}{Z_1 + Z_2} I$$

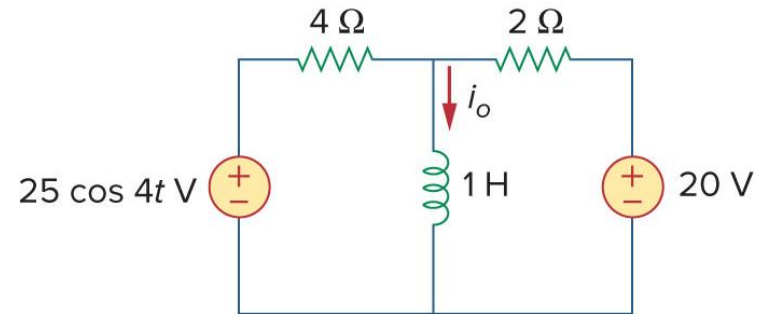
Topic 8 – AC analysis

- **Nodal** and **mesh analysis** are applied to AC circuits the same way as in DC circuits.
 - Nodal equations (KCL) and mesh equations (KVL) should be written for an AC circuit in phasor domain when it is operated in steady state sinusoidal conditions.
- **Superposition** can also be applied to AC circuits with multiple independent sources in phasor domain.
 - When having sources with **different frequencies only superposition** can be used to find the steady state response of the circuit.
 - A separate phasor circuit must be solved for each frequency independently.
 - The overall response is the sum of the time domain responses of all individual phasor circuits.
 - If the **frequencies are the same**, the sum of the phasor responses will be used as the final response and then transformed back to time domain if needed.

Topic 8 – AC analysis

Example of superposition in AC: Find i_o .

! **Note:** Superposition is this only option in this case, since there are two sources operating at different frequencies (AC source at 4 rad/s and DC source at 0 rad/s).



① I_o' due to AC voltage source (turn off DC voltage source)



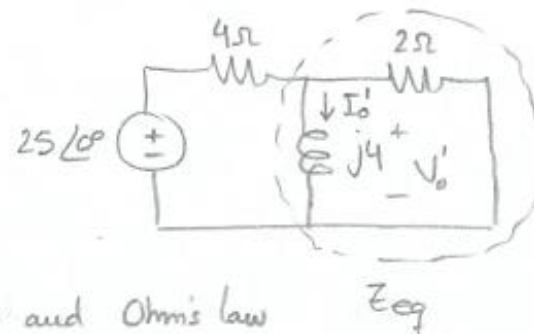
First, convert to phasor domain

$$V_1 = 25 \cos 4t \rightarrow V_1 = 25 \angle 0^\circ$$

$$Z_{4\Omega} = 4 \Omega$$

$$Z_{2\Omega} = 2 \Omega$$

$$Z_{1H} = j\omega L = j4 \times 1 = j4 \Omega$$

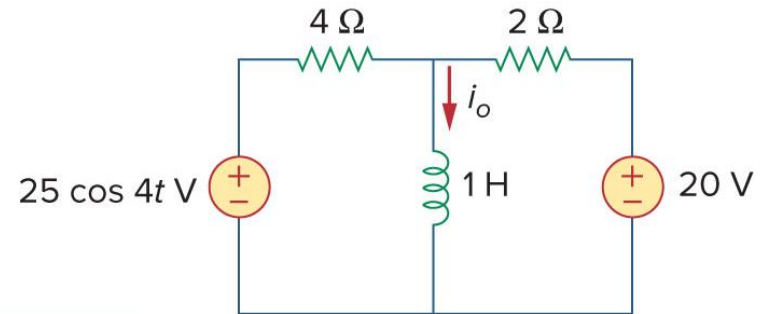


Then, calculate I_o using voltage divider and Ohm's law

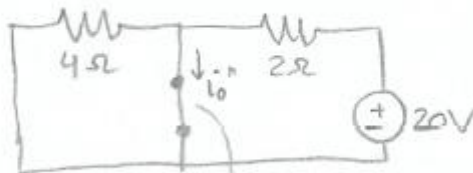
Topic 8 – AC analysis

Example of superposition in AC: Find i_o .

! **Note:** Superposition is this only option in this case, since there are two sources operating at different frequencies (AC source at 4 rad/s and DC source at 0 rad/s).



② i_o'' due to DC voltage source (turn off AC voltage source)



$$i_o'' = \frac{20V}{2\Omega} = 10 A$$

inductor behaves as short circuit in DC steady state

③ Final result

$$i_o = i_o' + i_o'' = 10 + 1.976 \cos(4t - 71.56^\circ)$$

↑
we can only add them as waveforms, not as phasors

Topic 8 – AC analysis

- The concept of **source transformation** is also applicable in the frequency domain.
- **Thevenin** and **Norton equivalent** circuits can also be used for AC circuits in phasor domain.
 - The Thevenin equivalent circuit consists of a voltage source V_{Th} in series with the Thevenin equivalent impedance Z_{Th} :
 - $V_{Th} = V_{oc}$: Open-circuit phasor voltage across the given terminals.
 - $Z_{Th} = Z_{eq}$: Equivalent impedance as seen from the given terminals.
 - The Norton equivalent circuit consists of a current source I_N in parallel with the Norton equivalent impedance Z_N :
 - $I_N = I_{sc}$: Short-circuit current through the given terminals.
 - $Z_N = Z_{Th} = Z_{eq}$: Equivalent impedance as seen from the given terminals.
 - **Thevenin-Norton transformation** is the same as in DC:

$$I_N = \frac{V_{Th}}{Z_{Th}} \text{ and } V_{Th} = Z_N I_N$$

Topic 8 – AC Op Amps

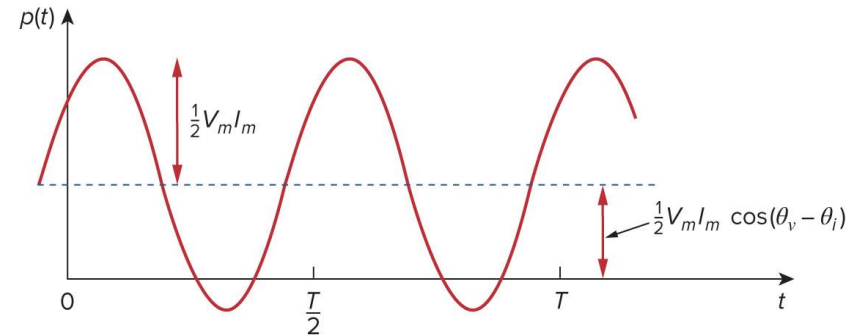
- AC Op Amp circuits are analyzed in the same way as DC circuits.
 - Transform an AC Op Amp circuit into **phasor domain**.
 - Apply **Nodal analysis** considering **Ideal** Op Amp.
 - **Input currents** to Op Amp are **zero**.
 - **Input voltage nodes** are **equal** (with negative feedback).
- It is possible to use the formulas for common DC Op Amps in AC mode.
 - Inverting amplifier: $\frac{V_o}{V_s} = -\frac{Z_f}{Z_1}$
 - Non-inverting amplifier: $\frac{V_o}{V_s} = 1 + \frac{Z_f}{Z_1}$
 - Summing amplifier: $V_o = -\left(\frac{Z_f}{Z_1}V_1 + \frac{Z_f}{Z_2}V_2 + \frac{Z_f}{Z_3}V_3\right)$

Topic 9 – AC power

Instantaneous power

- Instantaneous power is $p(t) = v(t)i(t)$.
- For **sinusoidal** voltage and current:

$$p(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$



Average power

- The average power P is the **average** of instantaneous power over **one period**.

$$P = \frac{1}{T} \int_0^T p(t) dt$$

- For **sinusoidal** voltage and current: $P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2} \text{Re}[\mathbf{V}\mathbf{I}^*]$

- Average power** absorbed by a **resistor** R is:

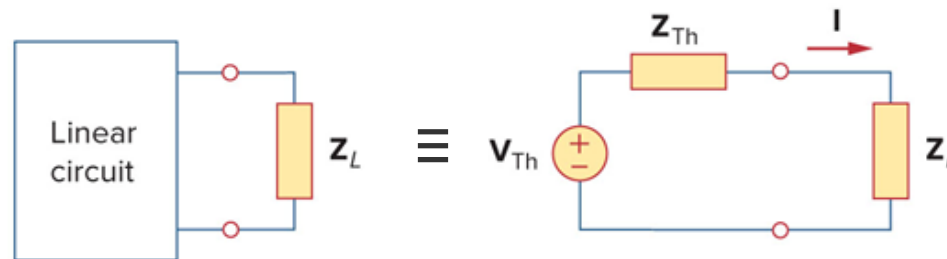
$$P = \frac{1}{2} V_m I_m = \frac{1}{2} R I_m^2 = \frac{1}{2} R |\mathbf{I}|^2 = \frac{1}{2} \frac{V_m^2}{R} = \frac{1}{2} \frac{|\mathbf{V}|^2}{R}$$

- Average power** absorbed by **inductor** and **capacitor** is **zero**.

Topic 9 – MAPT (maximum average power transfer)

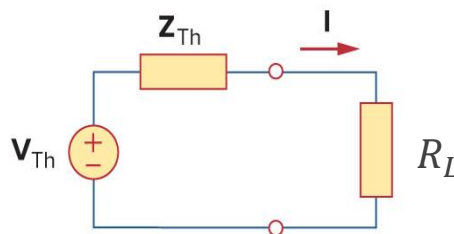
- Using **Thevenin** equivalent circuit, maximum average power is transferred to a complex load $\mathbf{Z}_L$ when:

$$\mathbf{Z}_L = \mathbf{Z}_{Th}^* \Rightarrow P_{\max} = \frac{|\mathbf{V}_{Th}|^2}{8R_{Th}}$$



- For pure resistive load R_L , maximum average power is transferred when:

$$R_L = \sqrt{R_{Th}^2 + X_{Th}^2} = |\mathbf{Z}_{Th}| \Rightarrow P_{\max} = \frac{1}{2} R_L |\mathbf{I}_L|^2 = \frac{1}{2} \frac{|\mathbf{V}_L|^2}{R_L}$$



Topic 9 - Effective or RMS value

- Effective or RMS value is a DC signal that can deliver the same average power to a resistor as the AC signal.
- Effective or RMS value take the form of the square root of the average (mean) of the square of the periodic signal.

$$X_{eff} = X_{rms} = \sqrt{\frac{1}{T} \int_0^T x(t)^2 dt}$$

- For sinusoidal voltage and current:

$$V_{eff} = V_{rms} = \frac{V_m}{\sqrt{2}}$$
$$I_{eff} = I_{rms} = \frac{I_m}{\sqrt{2}}$$

- Average power can be calculated using RMS values of current and voltage.

$$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$$

- Average power absorbed by a resistor R using RMS value:

$$P = V_{rms} I_{rms} = R I_{rms}^2 = \frac{V_{rms}^2}{R}$$

Questions?



